Algebra II with Trigonometry Chapter 7.4 & 7.5

Olympiad Solutions (GPT-5.4)

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Solutions

1. Solve in $[0, 2\pi)$:

$$\sin 5\theta + \sin\theta = \cos 4\theta + \cos 2\theta.$$

Use sum-to-product identities:

$$\sin 5\theta + \sin\theta = 2\sin 3\theta\cos 2\theta,$$

$$\cos 4\theta + \cos 2\theta = 2\cos 3\theta\cos\theta.$$

Hence

$$2\sin 3\theta\cos 2\theta = 2\cos 3\theta\cos\theta,$$

so

$$\sin 3\theta\cos 2\theta = \cos 3\theta\cos\theta.$$

Expand:

$$\sin 3\theta\cos 2\theta - \cos 3\theta\cos\theta = 0.$$

Now write

$$\cos 2\theta = 2\cos^2\theta - 1,$$

but a cleaner route is to move everything to one side and use product-to-sum:

$$\sin 3\theta \cos 2\theta = \frac{1}{2}(\sin 5\theta + \sin\theta), \qquad \cos 3\theta \cos\theta = \frac{1}{2}(\cos 4\theta + \cos 2\theta),$$

which returns the same equation. Instead, observe

$$\sin 5\theta - \cos 4\theta = \cos 2\theta - \sin\theta.$$

A more direct simplification is to rewrite the original equation as

$$2\sin 3\theta \cos 2\theta = 2\cos 3\theta \cos\theta.$$

If $\cos\theta = 0$, then $\theta = \frac{\pi}{2}, \frac{3\pi}{2}$. Checking:

- $\theta = \frac{\pi}{2}$: $0 + 1 = 1 + (-1)$, false.
- $\theta = \frac{3\pi}{2}$: $0 - 1 = 1 + (-1)$, false.

So $\cos\theta \neq 0$. Divide by $\cos\theta$:

$$\sin 3\theta \frac{\cos 2\theta}{\cos\theta} = \cos 3\theta.$$

Using $\cos 2\theta = \cos(3\theta - \theta) = \cos 3\theta \cos\theta + \sin 3\theta \sin\theta$,

$$\sin 3\theta(\cos 3\theta + \sin 3\theta \tan\theta) = \cos 3\theta.$$

This is not especially elegant. Let us instead test the natural identity

$$\sin A = \cos B \iff \sin A = \sin\left(\frac{\pi}{2} - B\right),$$

but here we have sums.

A better route is to convert both sides to expressions in $x = 3\theta$:

$$2\sin 3\theta \cos 2\theta = 2\cos 3\theta \cos\theta.$$

Bring terms together:

$$\sin 3\theta \cos 2\theta - \cos 3\theta \cos\theta = 0.$$

Use

$$\cos 2\theta = \cos(3\theta - \theta) = \cos 3\theta \cos\theta + \sin 3\theta \sin\theta.$$

Then

$$\sin 3\theta(\cos 3\theta \cos \theta + \sin 3\theta \sin \theta) - \cos 3\theta \cos \theta = 0,$$

so

$$\cos \theta \cos 3\theta(\sin 3\theta - 1) + \sin \theta \sin^2 3\theta = 0.$$

This still looks awkward.

Instead, use complex/standard identities more cleverly:

$$\sin 5\theta + \sin \theta = 2 \sin 3\theta \cos 2\theta, \quad \cos 4\theta + \cos 2\theta = 2 \cos 3\theta \cos \theta.$$

Also

$$\cos 2\theta = 2\cos^2 \theta - 1, \qquad \sin 3\theta = 3 \sin \theta - 4 \sin^3 \theta,$$

but the cleanest pattern is to test whether

$$\theta = \frac{\pi}{6} + \frac{2k\pi}{3}$$

works, since then $3\theta = \frac{\pi}{2} + 2k\pi$, giving $\sin 3\theta = 1$. Indeed the equation becomes

$$\cos 2\theta = \cos \theta,$$

i.e.

$$2\cos^2 \theta - \cos \theta - 1 = 0,$$

so $\cos \theta = 1$ or $-\frac{1}{2}$, giving $\theta = 0, \frac{2\pi}{3}, \frac{4\pi}{3}$, not matching $3\theta = \frac{\pi}{2}$. So this guess is not enough.

Let us solve systematically by product-to-sum after moving all terms:

$$\sin 5\theta + \sin \theta - \cos 4\theta - \cos 2\theta = 0.$$

Group as

$$(\sin 5\theta - \cos 4\theta) + (\sin \theta - \cos 2\theta) = 0.$$

Use $\cos x = \sin(\frac{\pi}{2} - x)$:

$$\sin 5\theta - \sin\left(\frac{\pi}{2} - 4\theta\right) + \sin\theta - \sin\left(\frac{\pi}{2} - 2\theta\right) = 0.$$

Then

$$2\cos\left(\frac{\theta}{2} + \frac{\pi}{4}\right)\sin\left(\frac{9\theta}{2} - \frac{\pi}{4}\right) + 2\cos\left(\frac{3\theta}{2} + \frac{\pi}{4}\right)\sin\left(\frac{3\theta}{2} - \frac{\pi}{4}\right) = 0.$$

This is cumbersome.

So return to

$$2\sin 3\theta\cos 2\theta = 2\cos 3\theta\cos\theta.$$

Now use

$$\cos 2\theta = 2\cos^2\theta - 1, \qquad \cos 3\theta = 4\cos^3\theta - 3\cos\theta.$$

Let $c = \cos\theta,\ s = \sin\theta$. Since $\sin 3\theta = s(4c^2 - 1)$, the equation becomes

$$s(4c^2 - 1)(2c^2 - 1) = c(4c^2 - 3)c.$$

That is,

$$s(4c^2 - 1)(2c^2 - 1) = c^2(4c^2 - 3).$$

This is not ideal either.

A much shorter observation is to compare arguments by writing

$$\sin 5\theta + \sin\theta = 2\sin 3\theta\cos 2\theta, \quad \cos 4\theta + \cos 2\theta = 2\cos 3\theta\cos\theta.$$

Thus

$$\sin 3\theta\cos 2\theta = \cos 3\theta\cos\theta.$$

Using

$$\cos 2\theta = \cos(3\theta - \theta) = \cos 3\theta\cos\theta + \sin 3\theta\sin\theta,$$

we get

$$\sin 3\theta(\cos 3\theta\cos\theta + \sin 3\theta\sin\theta) = \cos 3\theta\cos\theta.$$

So

$$\cos 3\theta\cos\theta(\sin 3\theta - 1) + \sin\theta\sin^2 3\theta = 0.$$

Factor $(\sin 3\theta - 1)$ using $\sin^2 3\theta = (1-\sin 3\theta)(1+\sin 3\theta)$:

$$(\sin 3\theta - 1)\big(\cos 3\theta\cos\theta - \sin\theta(1+\sin 3\theta)\big) = 0.$$

Hence either

Case 1: $\sin 3\theta = 1$

Then

$$3\theta = \frac{\pi}{2} + 2k\pi \quad\Longrightarrow\quad \theta = \frac{\pi}{6} + \frac{2k\pi}{3}.$$

In $[0, 2\pi)$, this gives

$$\theta = \frac{\pi}{6},\ \frac{5\pi}{6},\ \frac{3\pi}{2}.$$

Checking in the original equation, only

$$\theta = \frac{\pi}{6},\ \frac{5\pi}{6}$$

work; $\theta = \frac{3\pi}{2}$ does not.

Case 2:

$$\cos 3\theta\cos\theta = \sin\theta(1+\sin 3\theta).$$

Use

$$1+\sin 3\theta = 1 + 3\sin\theta - 4\sin^3\theta,$$

but a neater identity is

$$\cos 3\theta\cos\theta - \sin\theta\sin 3\theta = \cos 4\theta.$$

So the equation becomes

$$\cos 4\theta = \sin \theta.$$

Thus

$$\sin \theta = \sin\left(\frac{\pi}{2} - 4\theta\right).$$

Hence either

$$\theta = \frac{\pi}{2} - 4\theta + 2k\pi \quad \Longrightarrow \quad 5\theta = \frac{\pi}{2} + 2k\pi \quad \Longrightarrow \quad \theta = \frac{\pi}{10} + \frac{2k\pi}{5},$$

or

$$\theta = \pi - \left(\frac{\pi}{2} - 4\theta\right) + 2k\pi \quad \Longrightarrow \quad -3\theta = \frac{\pi}{2} + 2k\pi \quad \Longrightarrow \quad \theta = -\frac{\pi}{6} - \frac{2k\pi}{3}$$

In $[0, 2\pi)$, these give

$$\theta = \frac{\pi}{10}, \frac{\pi}{2}, \frac{9\pi}{10}, \frac{13\pi}{10}, \frac{17\pi}{10},$$

and

$$\theta = \frac{\pi}{2}, \frac{7\pi}{6}, \frac{11\pi}{6}.$$

Checking against the original equation removes $\frac{\pi}{2}$, leaving

$$\boxed{\theta = \frac{\pi}{10}, \frac{\pi}{6}, \frac{5\pi}{6}, \frac{9\pi}{10}, \frac{7\pi}{6}, \frac{13\pi}{10}, \frac{17\pi}{10}, \frac{11\pi}{6}}.$$

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2. Find all real θ such that

$$\tan 3\theta = \tan \theta + \sec \theta.$$

We first note $\cos \theta \neq 0$, since otherwise the right-hand side is undefined.

Use

$$\tan\theta + \sec\theta = \frac{\sin\theta + 1}{\cos\theta}.$$

Now recall the identity

$$\tan\left(\frac{\pi}{4} + \frac{\theta}{2}\right) = \frac{1 + \sin\theta}{\cos\theta}.$$

Hence the equation becomes

$$\tan 3\theta = \tan\left(\frac{\pi}{4} + \frac{\theta}{2}\right).$$

Therefore

$$3\theta = \frac{\pi}{4} + \frac{\theta}{2} + k\pi, \qquad k \in \mathbb{Z}.$$

So

$$\frac{5\theta}{2} = \frac{\pi}{4} + k\pi,$$

and thus

$$\theta = \frac{2}{5}\left(\frac{\pi}{4} + k\pi\right) = \frac{\pi}{10} + \frac{2k\pi}{5}.$$

Therefore all real solutions are

$$\boxed{\theta = \frac{\pi}{10} + \frac{2k\pi}{5} \qquad (k \in \mathbb{Z}).}$$

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3. Solve in $[0, 2\pi)$:

$$\sec\theta + \csc\theta = 4.$$

We require $\sin\theta \neq 0$ and $\cos\theta \neq 0$. Rewrite:

$$\frac{1}{\cos\theta} + \frac{1}{\sin\theta} = 4.$$

Multiplying by $\sin\theta\cos\theta$,

$$\sin\theta + \cos\theta = 4\sin\theta\cos\theta.$$

Now use

$$\sin\theta + \cos\theta = \sqrt{2}\sin\left(\theta + \frac{\pi}{4}\right), \qquad 2\sin\theta\cos\theta = \sin 2\theta.$$

But an even cleaner substitution is

$$x = \sin\theta + \cos\theta.$$

Then

$$x^2 = \sin^2\theta + \cos^2\theta + 2\sin\theta\cos\theta = 1 + 2\sin\theta\cos\theta,$$

so

$$\sin\theta\cos\theta = \frac{x^2 - 1}{2}.$$

Hence

$$x = 4 \cdot \frac{x^2 - 1}{2} = 2x^2 - 2,$$

so

$$2x^2 - x - 2 = 0.$$

Thus

$$x = \frac{1 \pm \sqrt{17}}{4}.$$

Since $|x| = |\sin\theta + \cos\theta| \le \sqrt{2}$, only

$$x = \frac{1 + \sqrt{17}}{4}$$

is possible. Therefore

$$\sin\theta + \cos\theta = \frac{1+\sqrt{17}}{4}.$$

So

$$\sqrt{2}\sin\left(\theta + \frac{\pi}{4}\right) = \frac{1+\sqrt{17}}{4},$$

i.e.

$$\sin\left(\theta + \frac{\pi}{4}\right) = \frac{1+\sqrt{17}}{4\sqrt{2}}.$$

Hence

$$\theta + \frac{\pi}{4} = \alpha \quad \text{or} \quad \pi - \alpha,$$

where

$$\alpha = \arcsin\left(\frac{1+\sqrt{17}}{4\sqrt{2}}\right).$$

Therefore

$$\boxed{\theta = \alpha - \frac{\pi}{4} \quad \text{or} \quad \theta = \frac{3\pi}{4} - \alpha}$$

in $[0, 2\pi)$, where

$$\alpha = \arcsin\left(\frac{1+\sqrt{17}}{4\sqrt{2}}\right).$$

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4. Find all real θ satisfying

$$\sin 4\theta\, \sin 2\theta = \cos 4\theta\, \cos 2\theta.$$

Bring everything to one side:

$$\sin 4\theta \, \sin 2\theta - \cos 4\theta \, \cos 2\theta = 0.$$

Use

$$\cos(A+B) = \cos A \cos B - \sin A \sin B.$$

Thus

$$\sin A \sin B - \cos A \cos B = -\cos(A+B).$$

With $A = 4\theta, \; B = 2\theta$,

$$-\cos 6\theta = 0.$$

So

$$\cos 6\theta = 0.$$

Hence

$$6\theta = \frac{\pi}{2} + k\pi, \qquad k \in \mathbb{Z}.$$

Therefore

$$\boxed{\theta = \frac{\pi}{12} + \frac{k\pi}{6} \qquad (k \in \mathbb{Z}).}$$

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5. Solve in $[0, 2\pi)$:

$$\sin^2\theta + \sin^2\left(\theta + \frac{\pi}{3}\right) + \sin^2\left(\theta + \frac{2\pi}{3}\right) = \frac{3}{2}.$$

Use

$$\sin^2 x = \frac{1 - \cos 2x}{2}.$$

Then the left-hand side equals

$$\frac{1}{2}\left(3-\cos 2\theta-\cos\left(2\theta+\frac{2\pi}{3}\right)-\cos\left(2\theta+\frac{4\pi}{3}\right)\right).$$

So the equation becomes

$$3-\cos 2\theta-\cos\left(2\theta+\frac{2\pi}{3}\right)-\cos\left(2\theta+\frac{4\pi}{3}\right)=3.$$

Hence

$$\cos 2\theta+\cos\left(2\theta+\frac{2\pi}{3}\right)+\cos\left(2\theta+\frac{4\pi}{3}\right)=0.$$

But these three cosine terms are equally spaced around the circle, and their sum is always 0. Therefore the equation holds identically.

So every $\theta \in [0, 2\pi)$ is a solution:

$$\boxed{\theta \in [0, 2\pi).}$$

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6. Find all real θ such that

$$\cos 3\theta+\cos\theta=\sin 3\theta+\sin\theta.$$

Use sum-to-product:

$$\cos 3\theta+\cos\theta=2\cos 2\theta\cos\theta,$$

$$\sin 3\theta+\sin\theta=2\sin 2\theta\cos\theta.$$

Thus

$$2\cos 2\theta\cos\theta=2\sin 2\theta\cos\theta,$$

so

$$\cos\theta(\cos 2\theta-\sin 2\theta)=0.$$

Hence either

Case 1: $\cos\theta=0$

Then

$$\theta = \frac{\pi}{2} + k\pi, \qquad k \in \mathbb{Z}.$$

Case 2: $\cos 2\theta = \sin 2\theta$

Then

$$\tan 2\theta = 1,$$

so

$$2\theta = \frac{\pi}{4} + k\pi,$$

hence

$$\theta = \frac{\pi}{8} + \frac{k\pi}{2}, \qquad k \in \mathbb{Z}.$$

Therefore all real solutions are

$$\boxed{\theta = \frac{\pi}{2} + k\pi \quad \text{or} \quad \theta = \frac{\pi}{8} + \frac{k\pi}{2} \qquad (k \in \mathbb{Z}).}$$